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Project Title:	Interaction Between IGF-1 and Integrin Signaling in the Response of Bone to Mechanical Load		
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	POSTDOCTORAL FELLOWSHIP Bone loss that results from the lack of mechanical stimulation in the weightlessness of spaceflight is a significant concern for prolonged space exploration missions. However, the mechanisms of bone loss in microgravity have not been fully understood. Evidence supports a role for insulin-like growth factor type 1 (IGF-1) signaling in mediating skeletal response to mechanical load. Skeletal unloading by hindlimb elevation, a ground-based model for the unloading of spaceflight, causes bone loss due to reduced bone formation. IGF-1 is a potent growth factor in bone and is produced in osteocytes and mature osteoblasts after mechanical load. This IGF-1 may produce new osteoblasts and differentiate them into mature osteoblasts to form new bone. Furthermore, skeletal unloading leads to resistance to the anabolic effect of IGF-1. IGF-1 fails to increase bone formation and osteoblast proliferation in the unloaded bone. This failure to respond to IGF-1 is due to a failure of IGF-I to stimulate the phosphorylation of the IGF-1 receptor (IGF-1R), although		

Task Description:	IGF-1R levels and binding of IGF-1 to its receptor are not changed. Skeletal unloading decreases integrin expression in osteoblasts, and this impairment of expression is restored by reloading the bones or the osteoblasts in parallel with the recovery of IGF-1 responsiveness. Knockdown of beta1 and beta3 integrins blocks the ability of IGF-1 to activate its receptor and inhibits its activation by mechanical loading. We will use axial cyclic loading of the tibia as well as reloading following skeletal unloading in mice lacking IGF-1R or beta1 and beta3 integrins to determine the importance of these molecules in the response of bone to mechanical load. We will also use pulsatile fluid flow, a mechanical loading modality in vitro, to determine the molecular mechanism by which integrin signaling by mechanical load enhances IGF-1 signaling. If this hypothesis proves to be true, efforts to maintain the integrin signaling and IGF-1 signaling will be sought to prevent bone loss associated with skeletal unloading. Key findings: Mice lacking the IGF-1R in mature osteoblasts and osteocytes were subjected to skeletal unloading using hindlimb unloading and reloading either after unloading or with cyclic axial loading of the right tibia during hindlimb unloading. µCT analysis showed that the response of bone volume fraction (BV/TV) of the trabecular bone in the distal femur to unloading and reloading is similar in IGF-1R knockout (KO) mice to that in control mice at baseline and after two-week unloading and two weeks of reloading after the two-week unloading. Moreover, the response of the trabecular bone and cortical bone in the tibia to cyclic loading was comparable in KO mice to control mice. To further investigate response of the KO mice to reloading, bone histomorphometric analysis was performed. Bone formation rate, mineral apposition rare and mineralizing surface were markedly reduced in the periosteal cortical bone of the tibia of the reloaded KO mice after unloading compare to those in the control mice. Impact of key findings: The above findings indicate that IGF-1R plays a significant role in response to reloading after unloading, at least, in the cortical bone. Bone histomorphometric analysis in the trabecular bone is being performed to elucidate the underlying cellular mechanisms. We postulate that bone resorption is also reduced during the reloading after the unloading perhaps secondary to the reduction in bone formation. Proposed research plan: We will test mice deleted for integrin beta1 and beta3 from mature osteobasts and osteocytes to determine whether the skeletal response to mechanical load and IGF-1 infusion is impaired. We will also knock down or inhibit selected components of the integrin/IGF-1R complex to determine whether this blocks IGF-1R activation and/or the formation of the integrin/IGF-1R complex during pulsatile fluid flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bone loss is a serious problem in immobilized people due to injury or surgery on earth. A reduction in bone mass raises a risk of bone fracture. It takes a long time for those patients to recover bone strength in rehabilitation. The mechanisms of bone gain after weight-bearing activity following long-term bed rest have not been fully elucidated as well as those of bone loss during immobilization. Insulin-like growth factor 1 (IGF-1) signaling plays a vital role in bone remodeling. IGF-1 production is increased in osteocytes and osteoblasts after mechanical load. Skeletal unloading by hindlimb elevation, an animal model for immobilization on earth as well as a ground-based model for the unloading of spaceflight, causes a decrease in bone mass associated with resistance to the anabolic effect of IGF-1. Thus, we postulated that IGF-1 signaling mediates the skeletal response to unloading and reloading. To determine whether the IGF-1 receptor (IGF-1R) in mature osteoblasts and osteocytes has a role in the skeletal response to unloading and reloading, we evaluated mice in which the IGF-1R was deleted with osteocalcin-driven cre recombinase. These mice were subjected to skeletal unloading using hindlimb unloading and reloading after unloading. We found that IGF-1 signaling was critical for bone formation after reloading following skeletal unloading. Bone formation rate, mineral apposition rate and mineralizing surface were decreased in the periosteal cortical bone of IGF-1R deficient mice compare to control mice. These results suggest that IGF-1 signaling is necessary for promoting bone formation after weight-bearing activity following immobilization on earth and that efforts to maintain and potentiate the IGF-1 signaling will be sought to enhance bone formation following long-term bed rest.
Task Progress:	Bone loss is a serious health risk for prolonged space missions. It takes a long time for astronauts to recover bone strength in normal gravity after the microgravity of space. The mechanisms of bone gain after weight-bearing reloading following space flights have not been fully understood as well as those of bone loss during the microgravity of space. Insulin-like growth factor 1 (IGF-1) signaling plays a vital role in bone remodeling. IGF-1 production is increased in osteocytes and osteoblasts after mechanical load. Skeletal unloading by hindlimb elevation, a ground-based model for the unloading of spaceflight, causes a decrease in bone mass associated with resistance to the anabolic effect of IGF-1. Thus, we postulated that IGF-1 signaling mediates the skeletal response to unloading and reloading. To determine whether the IGF-1 receptor (IGF-1R) in mature osteoblasts and osteocytes has a role in the skeletal response to unloading and reloading, we evaluated mice in which the IGF-1R was deleted with osteocalcin-driven cre recombinase. These mice were subjected to skeletal unloading using hindlimb unloading and reloading after unloading. We found that IGF-1 signaling was critical for bone formation after reloading following skeletal unloading. Bone formation rate, mineral apposition rate and mineralizing surface were reduced in the periosteal cortical bone of IGF-1R deficient mice compare to control mice. These results suggest that IGF-1 signaling is required for accelerating bone formation after returning to normal gravity and that attempts to maintain and promote the IGF-1 signaling will be sought to augment bone formation following long-term space missions.
Bibliography Type:	Description: (Last Updated: 07/29/2020)
Abstracts for Journals and Proceedings	Kubota T, Elalieh HZ, Wang Y, Bikle DD. "IGF-1 receptor in mature osteoblasts and osteocytes is involved in skeletal unloading induced bone loss but not in reloading induced bone acquisition." 32nd Annual Meeting of the American Society for Bone and Mineral Research, Toronto, ON, Canada, October 15-19, 2010. J Bone Miner Res 2010 Sep;25(Suppl 1):1065. Available at: http://www.asbmr.org/Meetings/AnnualMeeting/AbstractDetail.aspx?aid=f16a1ce9-b9c2-46ea-8bb0-285522c55adc ; accessed 12/9/2011. , Oct-2010
Abstracts for Journals and Proceedings	Kubota T, Long RK, Elalieh HZ, Bikle DD. "Interaction between IGF-1 and integrin signaling in the response of bone to mechanical load." NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. NASA Human Research Program Investigators' Workshop, Abstract Book, February 2010. , Feb-2010